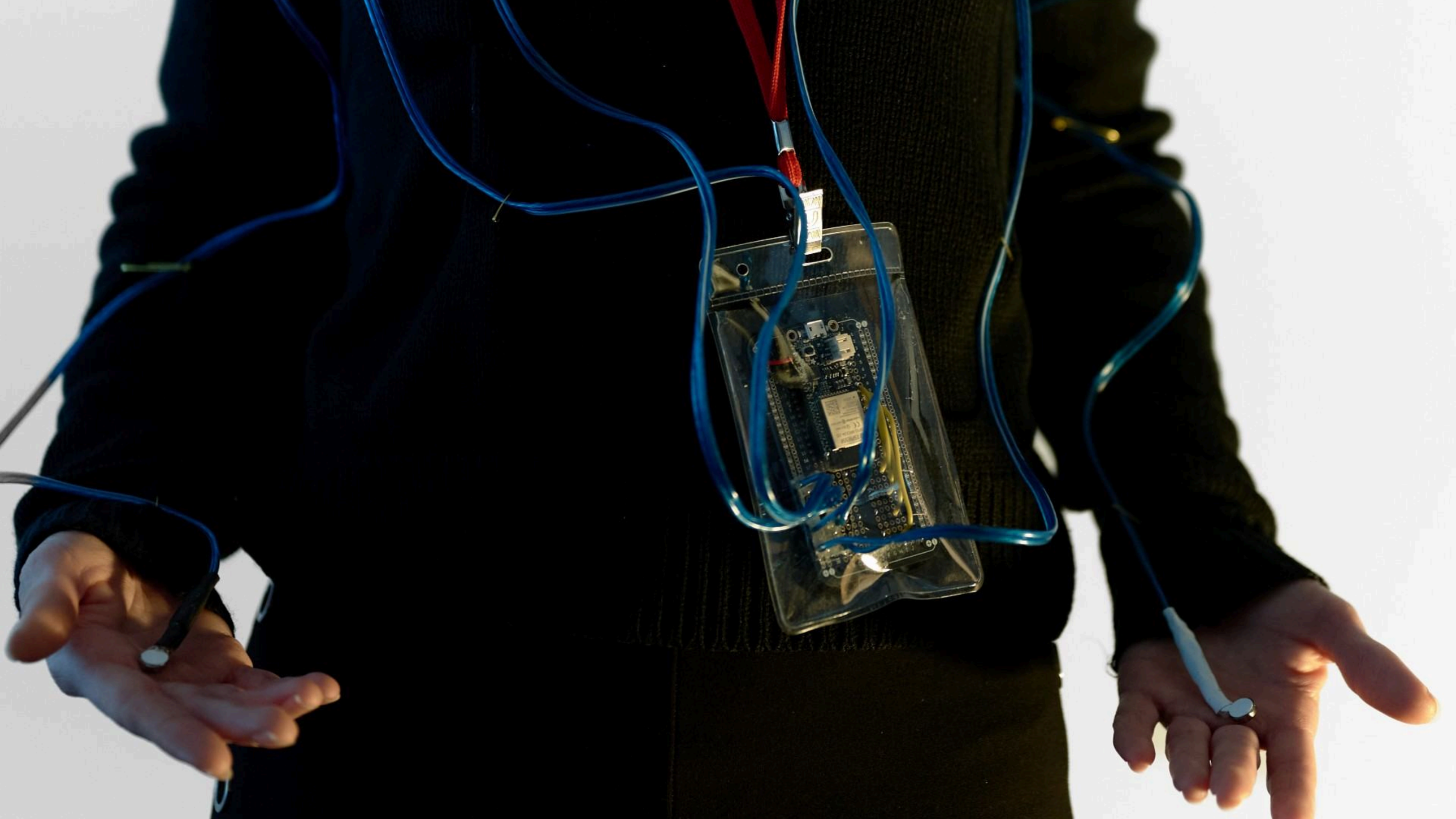
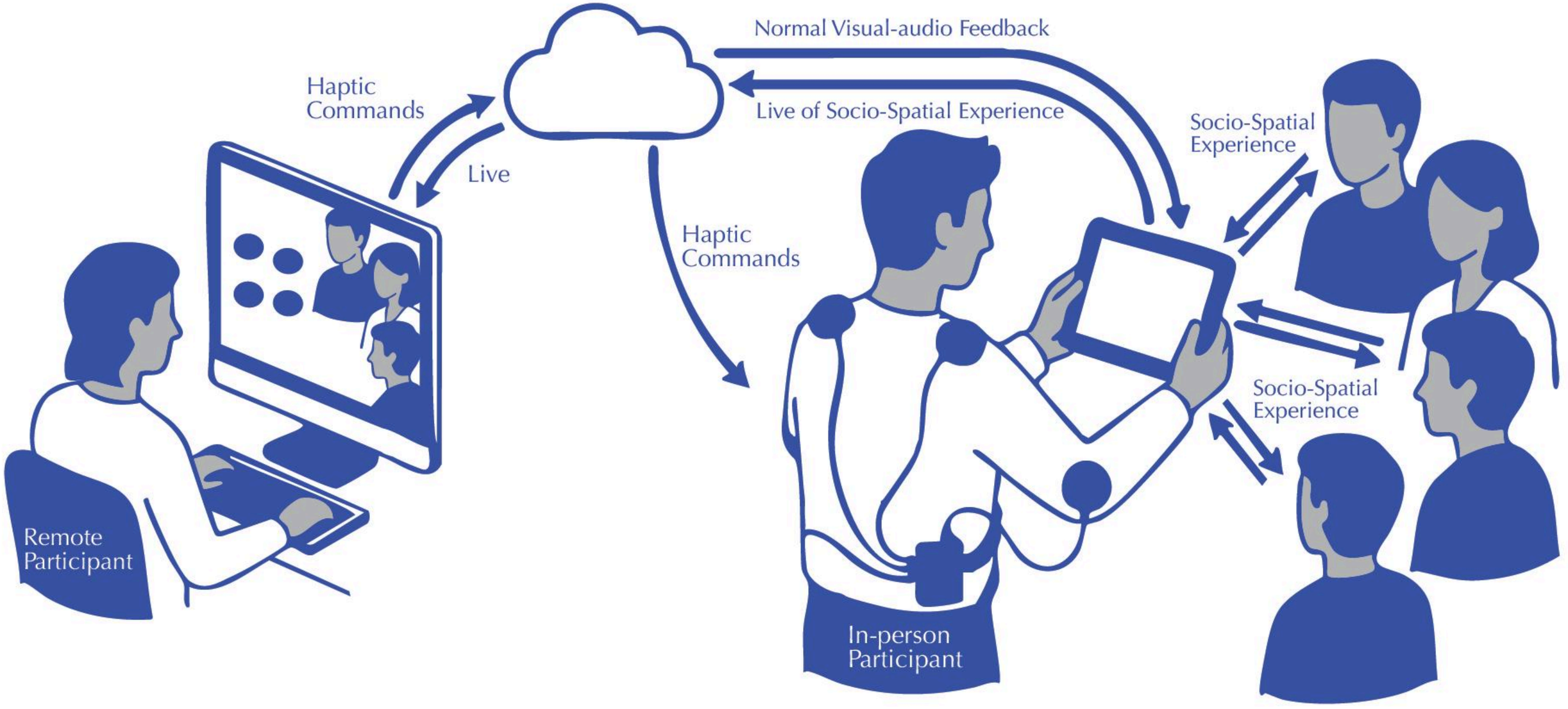
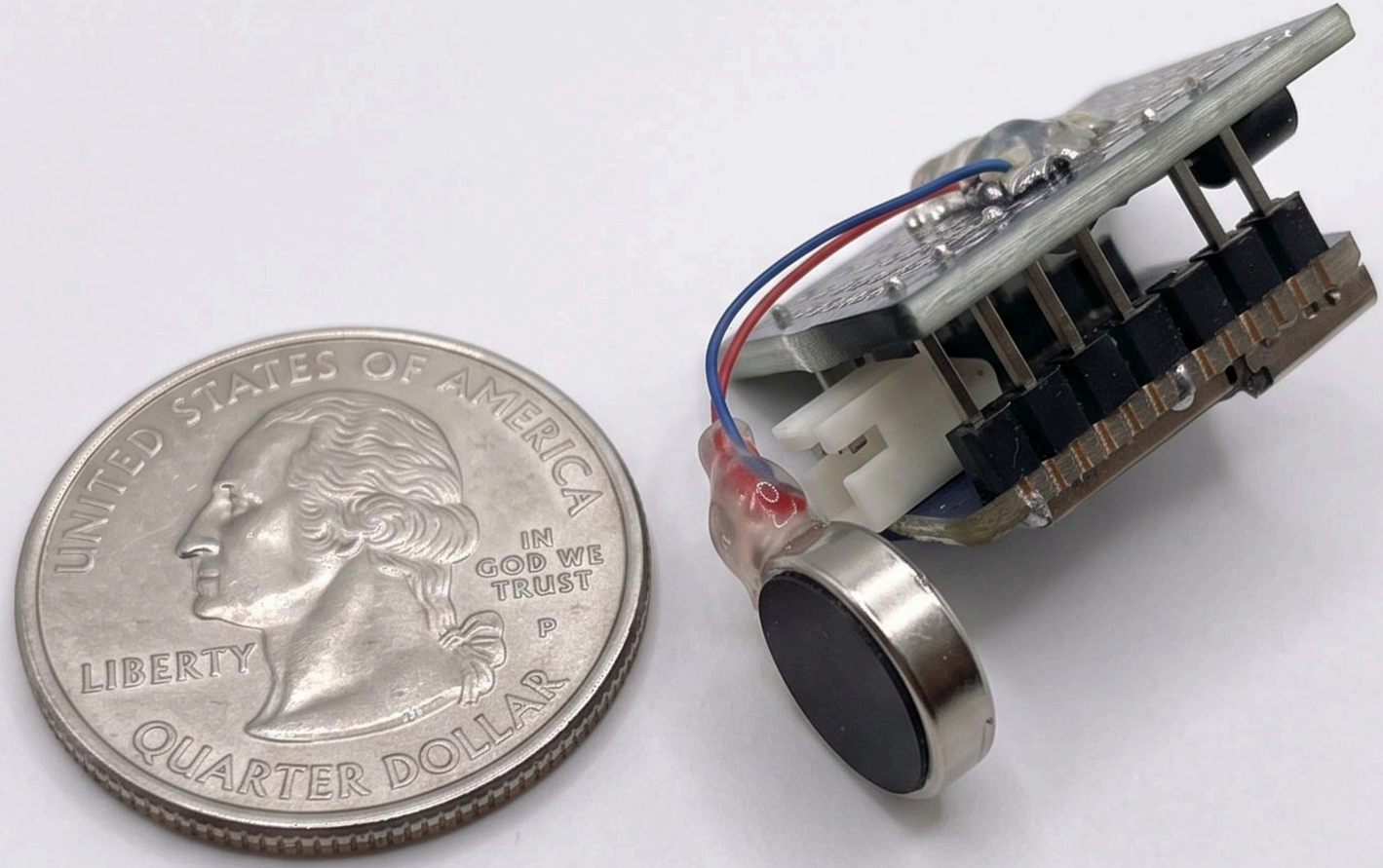
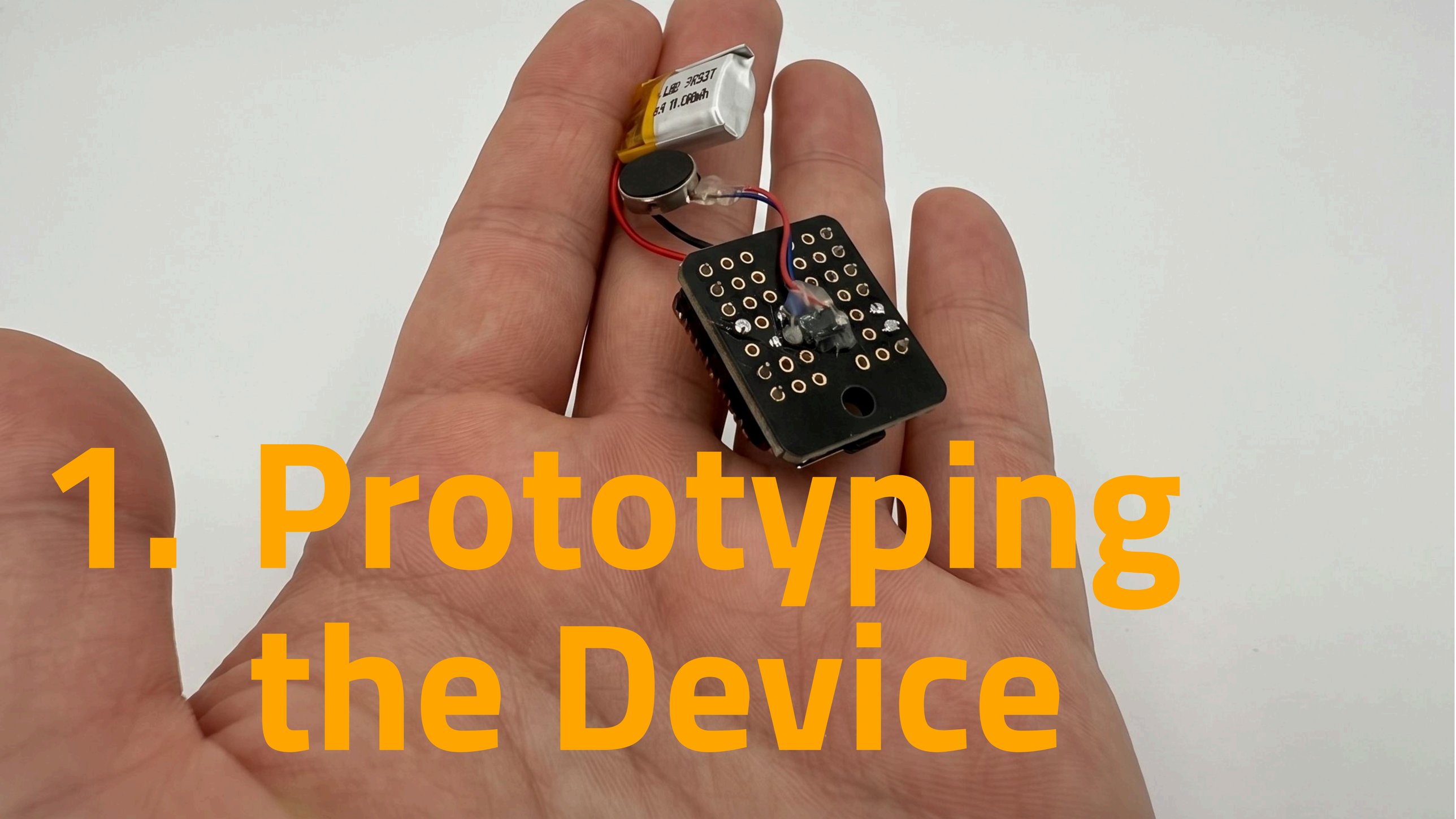
Sang Leigh, Cornell University
Sara Falcone, Pace University
Research sponsored by Sloan Foundation

Lending a Body Workshop



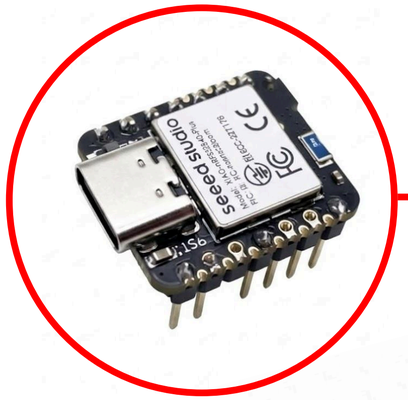

Normal Visual-audio Feedback
Haptic Commands
Live of Socio-Spatial Experience
Live
Socio-Spatial Experience
Haptic Commands
Socio-Spatial Experience
Remote Participant
In-person Participant


UNITED STATES OF AMERICA
IN GOD WE TRUST
LIBERTY
P
QUARTER DOLLAR



1. Prototyping the Device

MATERIALS



1. Seeed Studio XIAO nRF52840



2. JST PH 2.0mm Pitch 2-Pin



3. 2N7000 Mosfet Transistor

4. Protoboard

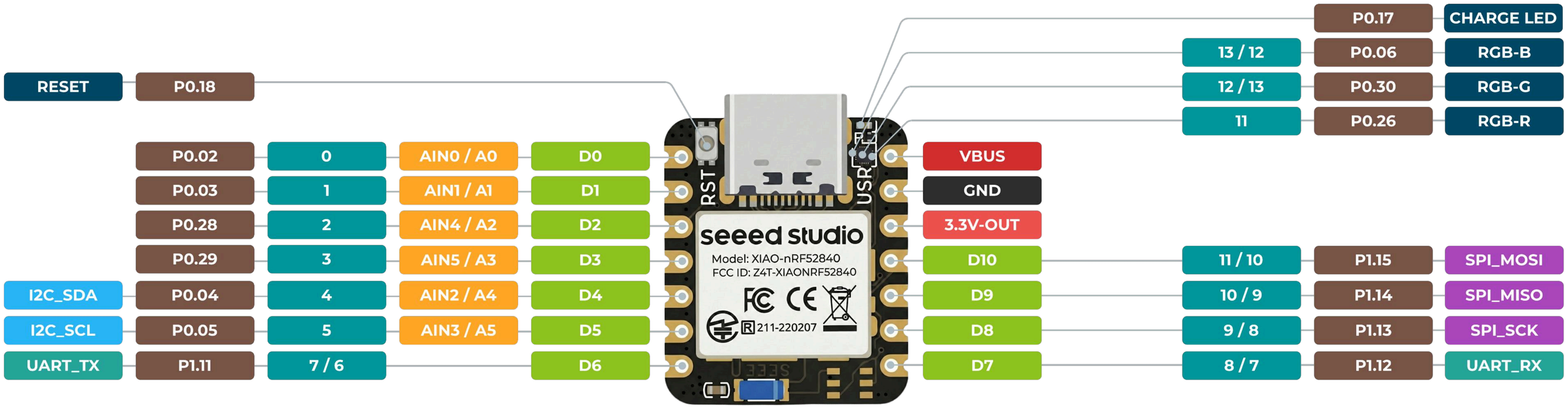
5. LRA vibration motor

6. 50mah Li-Po Battery
(any capacity works.
check connector polarity)



MICROCONTROLLER INFORMATION

https://wiki.seeedstudio.com/XIAO_BLE/

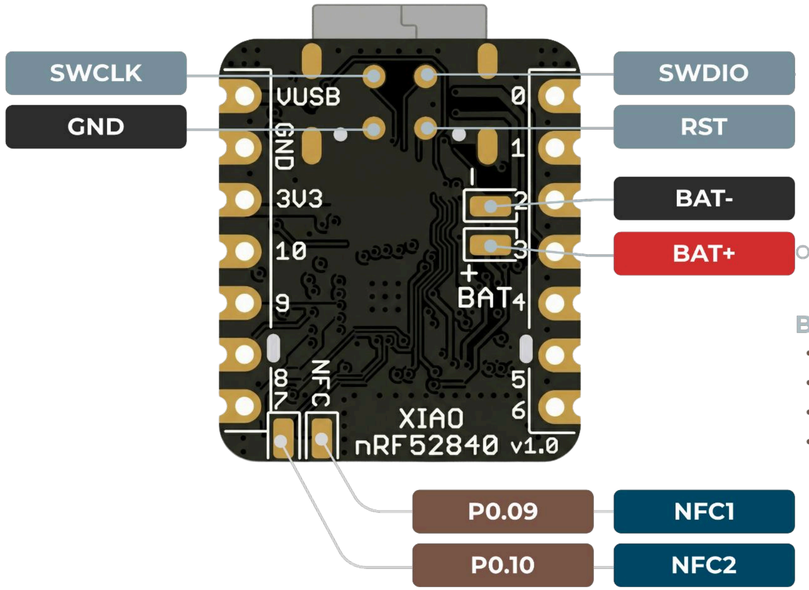


POWER | GND | DIGITAL GPIO | ADC INPUT | ARDUINO NAME | PIN NAME | SPI | UART | I2C | PERIPHERAL | SYSTEM

Pin numbering varies by Board Package. Format is displayed as: Mbed Core / Non-Mbed Core

MICROCONTROLLER INFORMATION

https://wiki.seeedstudio.com/XIAO_BLE/



Battery Management Logic

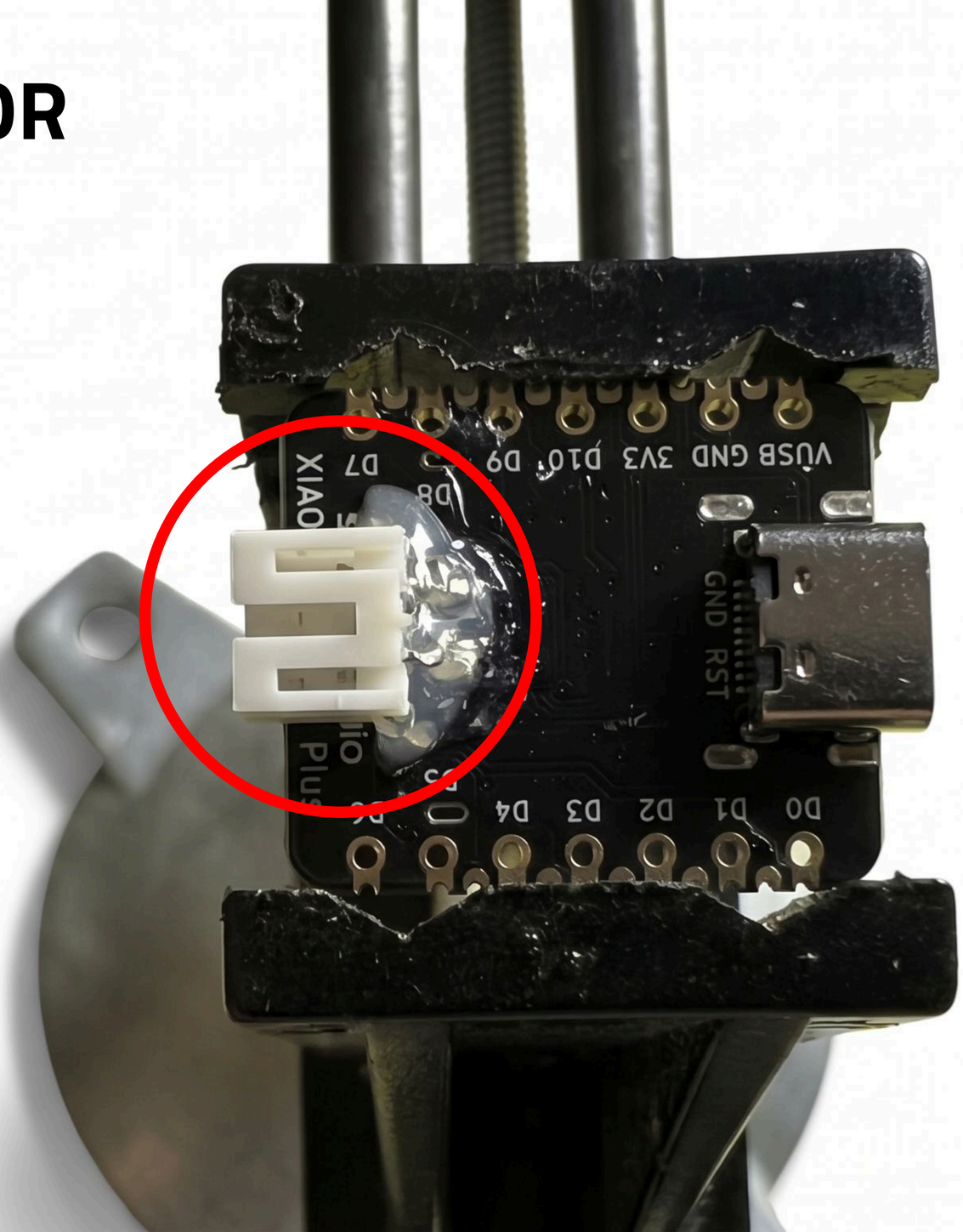
- **P0.31** / **35** (VBAT_ADC): Analog Input
- **P0.14** / **14** (READ_EN): Low = Enable Reading / High = Power Save
- **P0.13** / **22** (CHG_CTRL): High = Fast Charge / Low = Std Charge
- **P0.17** (CHG_STAT): Low = Charging / High = Full/Idle

POWER | GND | PIN NAME | SYSTEM | PERIPHERAL

1. BATTERY CONNECTOR

Please note the orientation.

The +/- polarity should match the battery’s



2. HEADER PINS



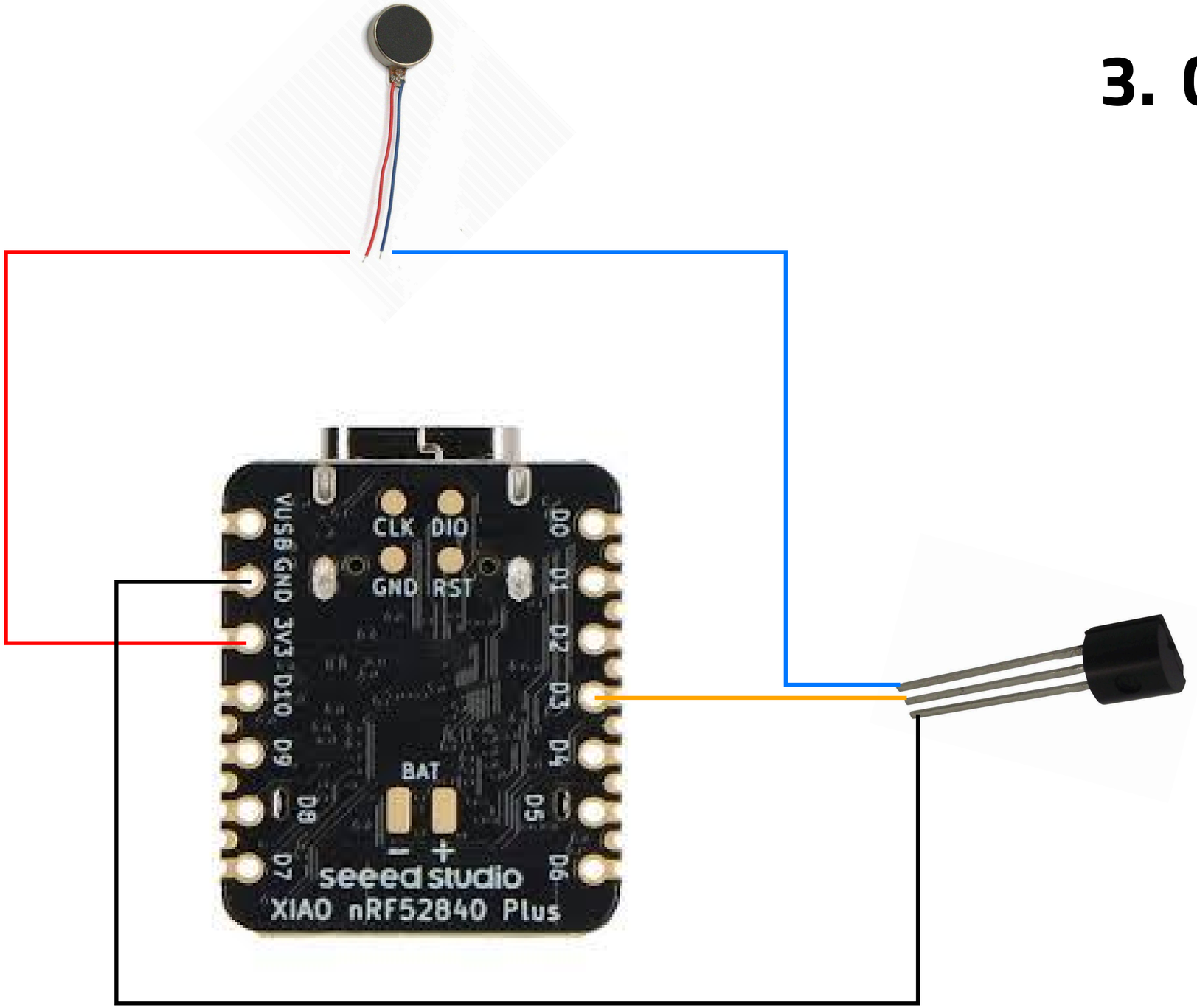


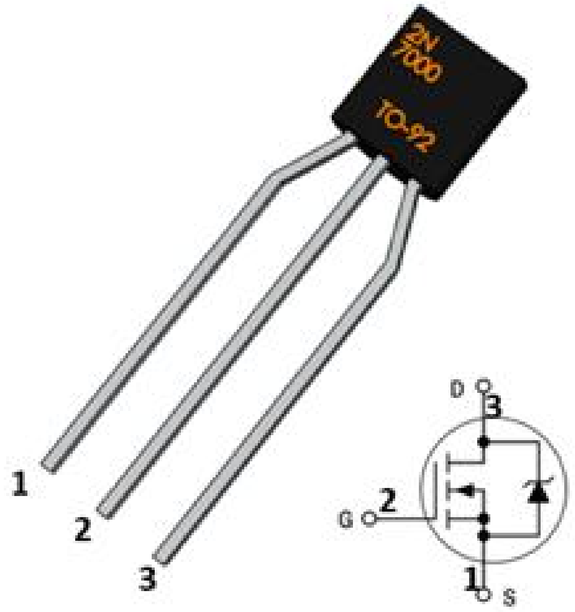


Pins used: GND, Vcc, and D3

Pins to remove: D2

3. CIRCUIT DIAGRAM





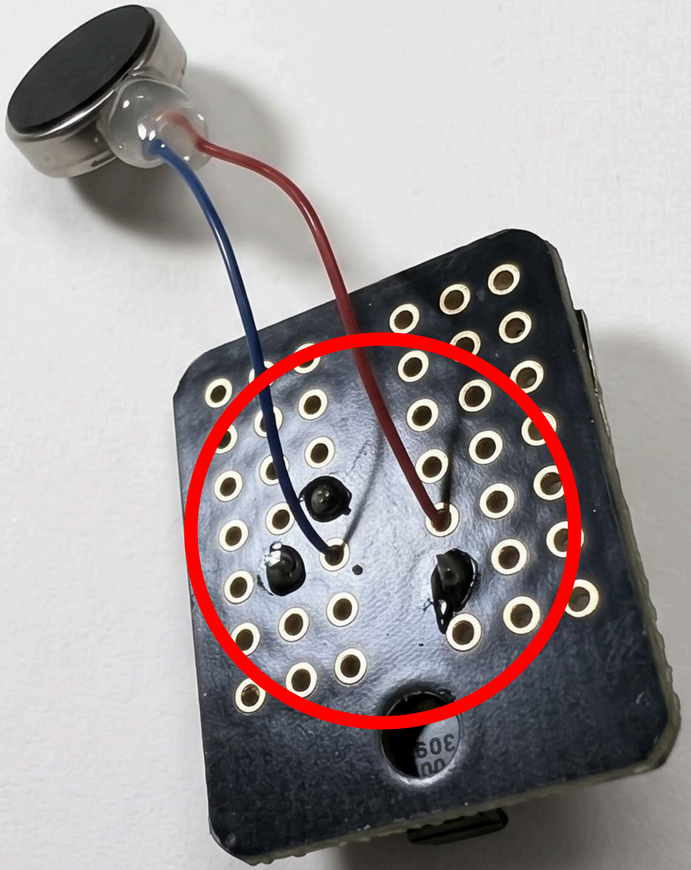
2N7000	
1	Source
2	Gate
3	Drain

4. PREPARING MOSFET AND PROTOBOARD



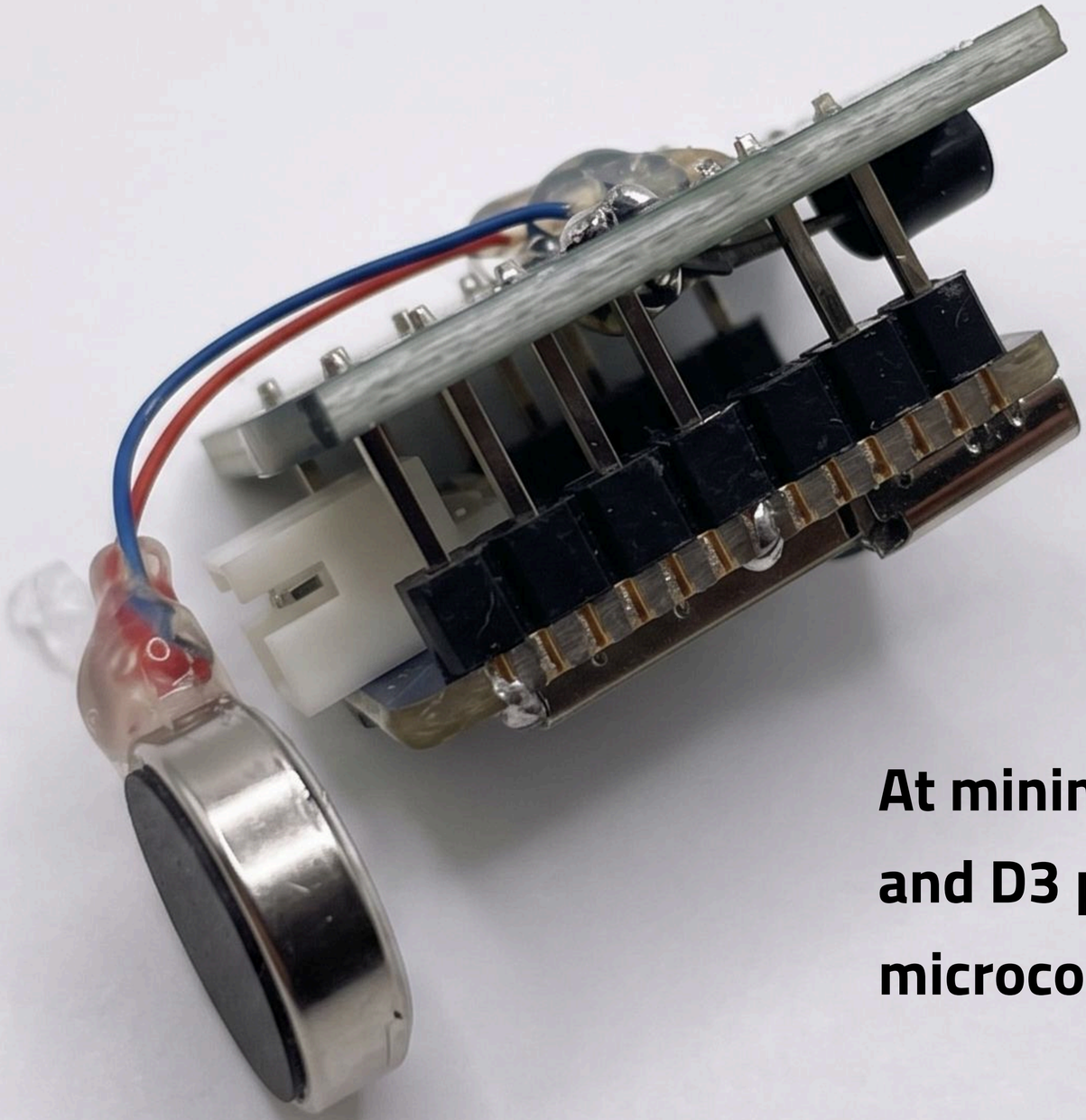
The Gate (middle) pin is bent to avoid contact with others.

5. CONNECTING MOTOR

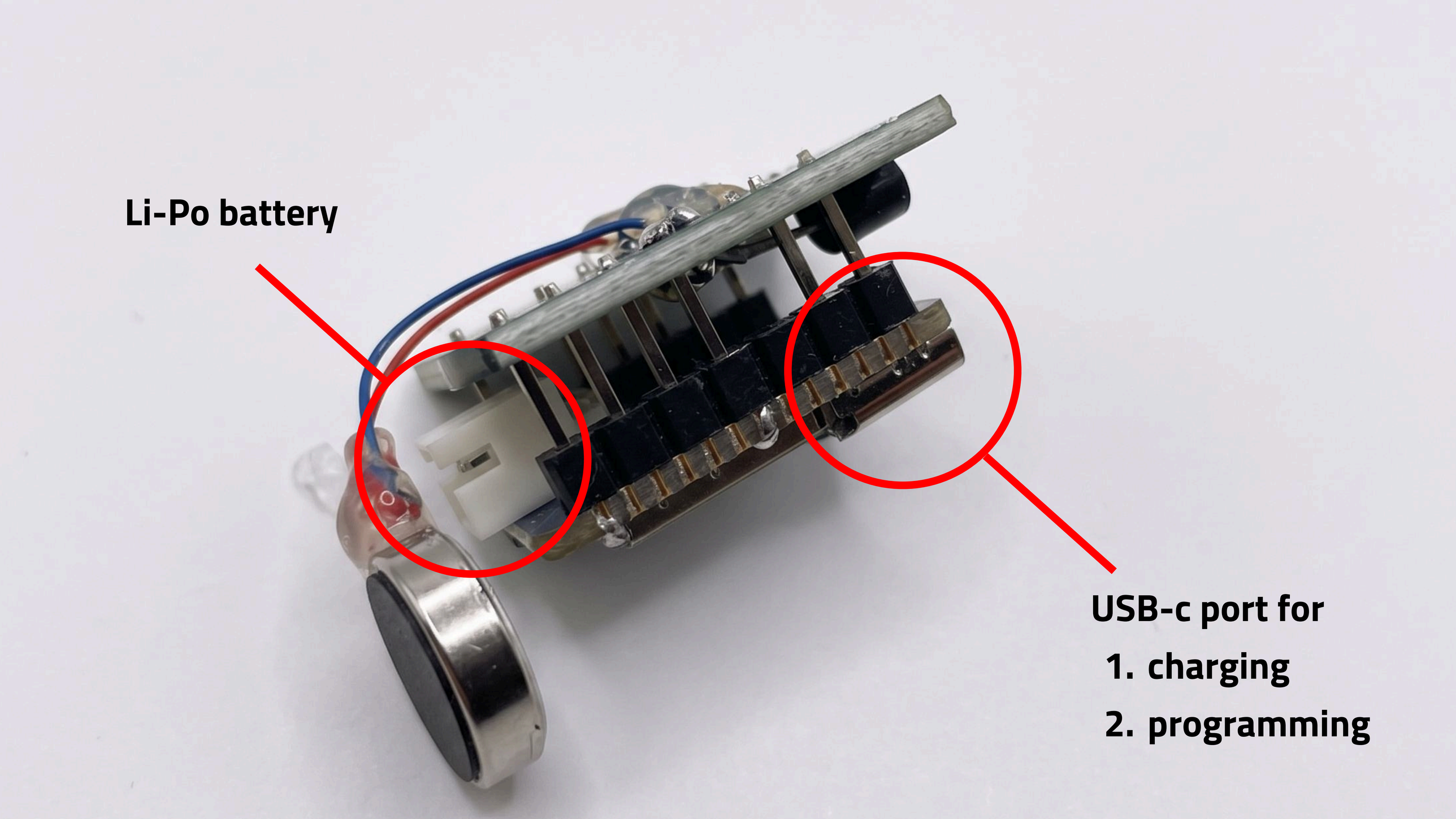




6. SOLDERING THE BOARD ON



At minimum, solder GND, Vcc, and D3 pins between the microcontroller and protoboard


Li-Po battery
USB-c port for
1. charging
2. programming



2. Running the Device

design-factory-junebug

Private

Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Go to file T Add file Code

sangww init	c34ee90 · yesterday	1 Commit
arduino/xiao_haptic_ble	init	yesterday
public	init	yesterday
.env	init	yesterday
.gitignore	init	yesterday
generate-cert.js	init	yesterday
package-lock.json	init	yesterday
package.json	init	yesterday
readme.md	init	yesterday
server.js	init	yesterday

README

Node Package Manager

NVM is highly recommended for managing multiple versions of Node.

Mac instruction

First, we need to install Homebrew:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

About

No description, website, or topics provided.

Readme
Activity
0 stars
0 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

sangww Sang

Languages

HTML 53.8% JavaScript 43.7%
C++ 2.5%

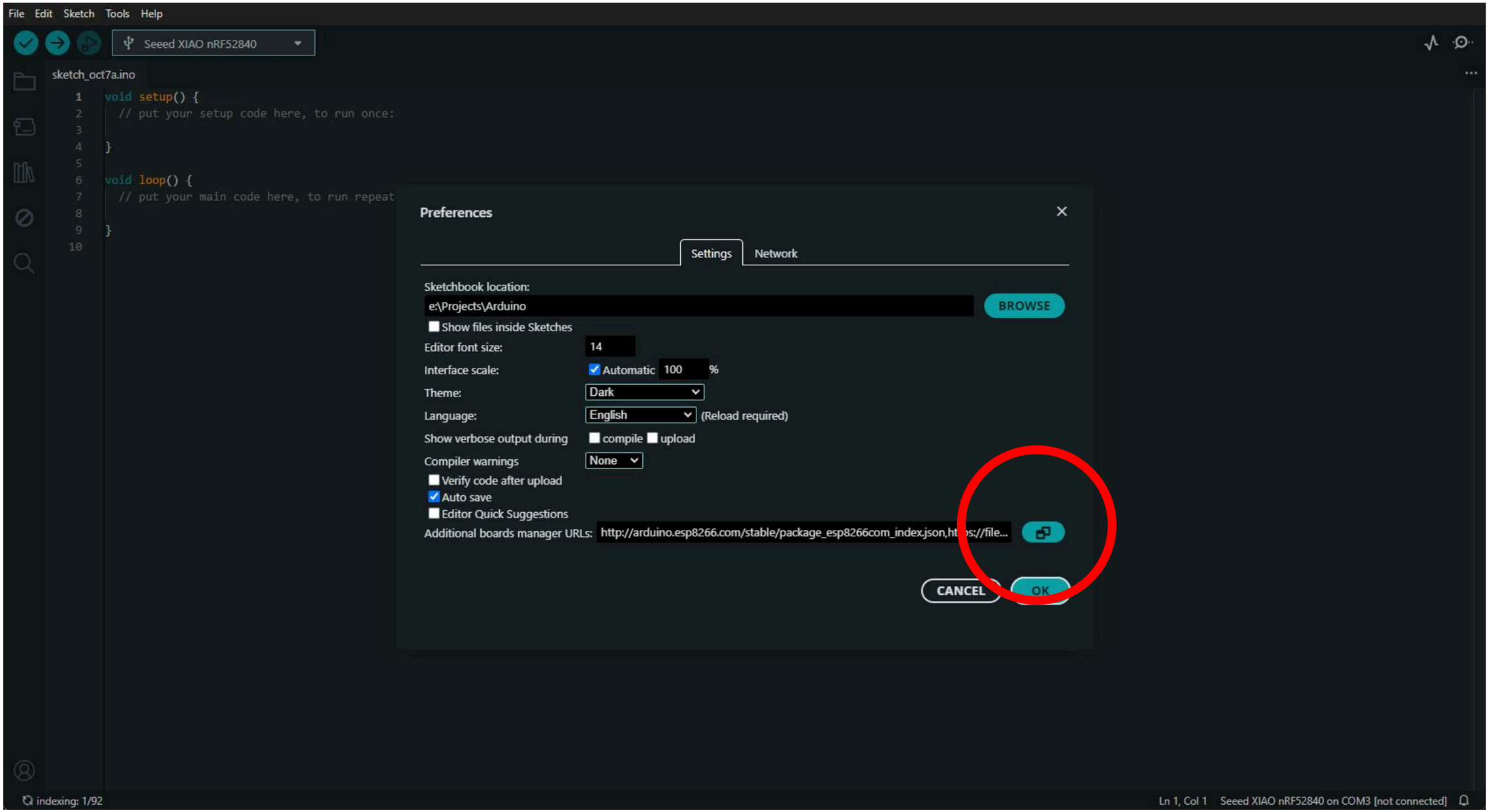
Suggested workflows

Based on your tech stack

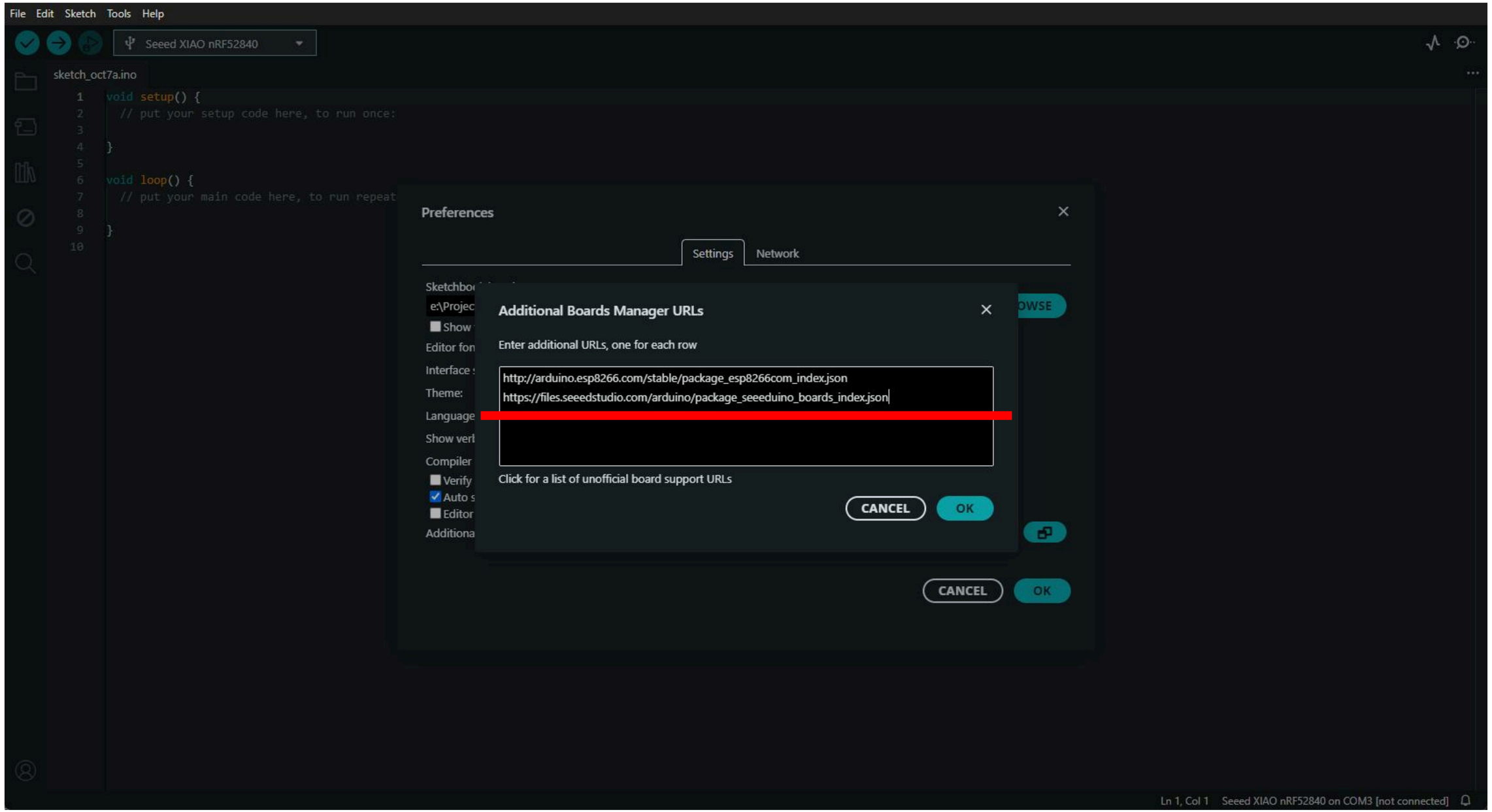
Jekyll using Docker Configure

https://github.com/sangww/design-factory-junebug

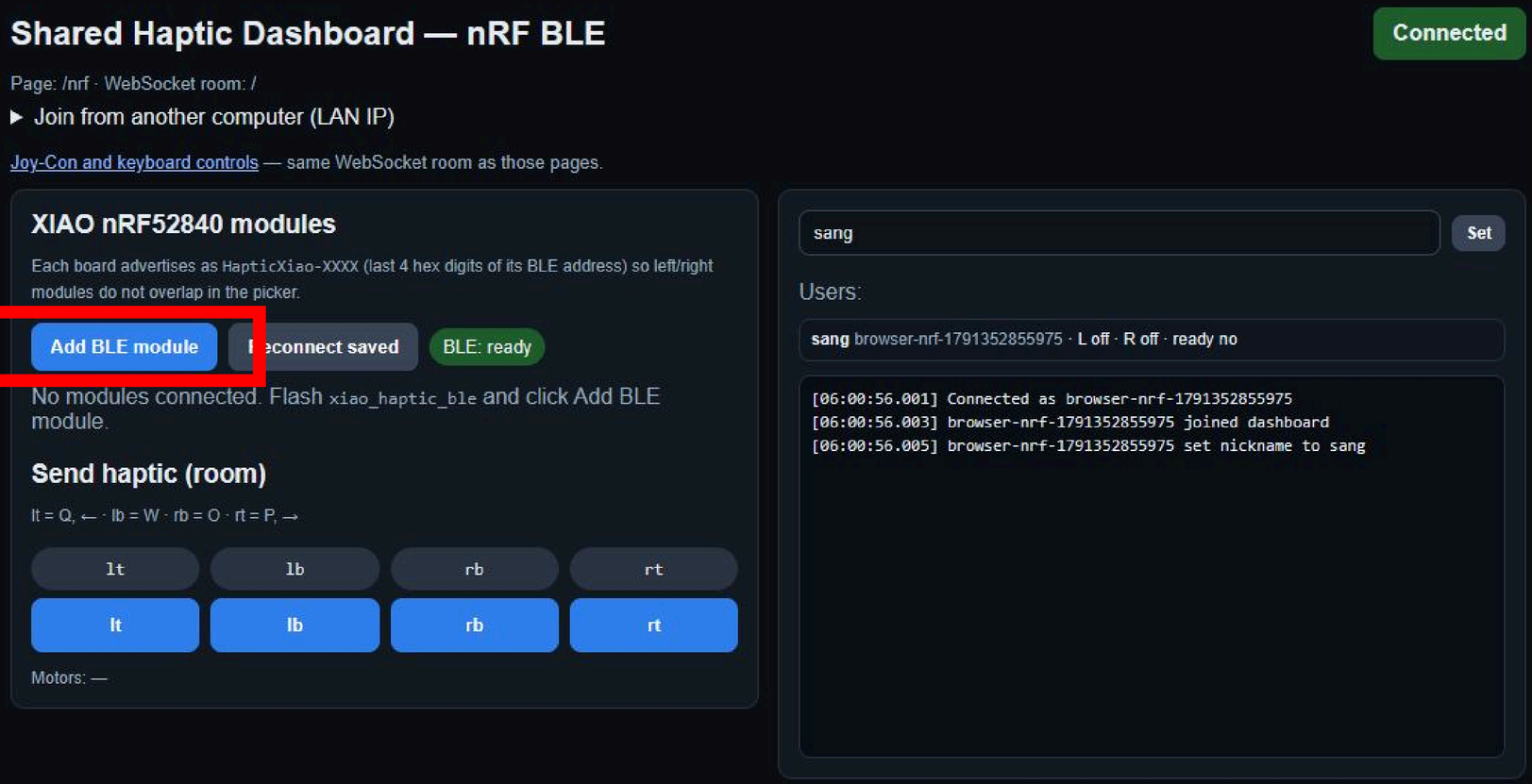
READ THE INSTURCTION HERE!



Adding the Microcontroller to Arduino IDE

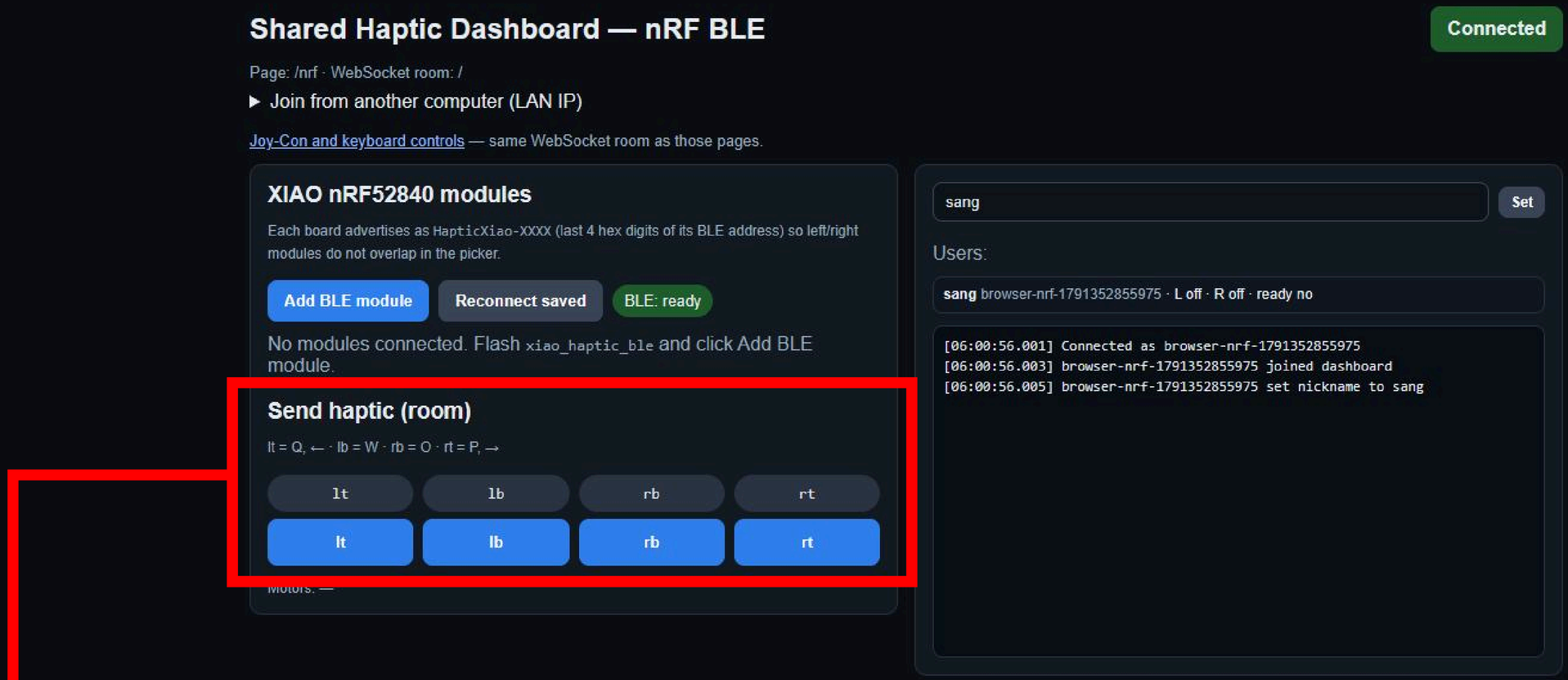


Adding the Microcontroller to Arduino IDE



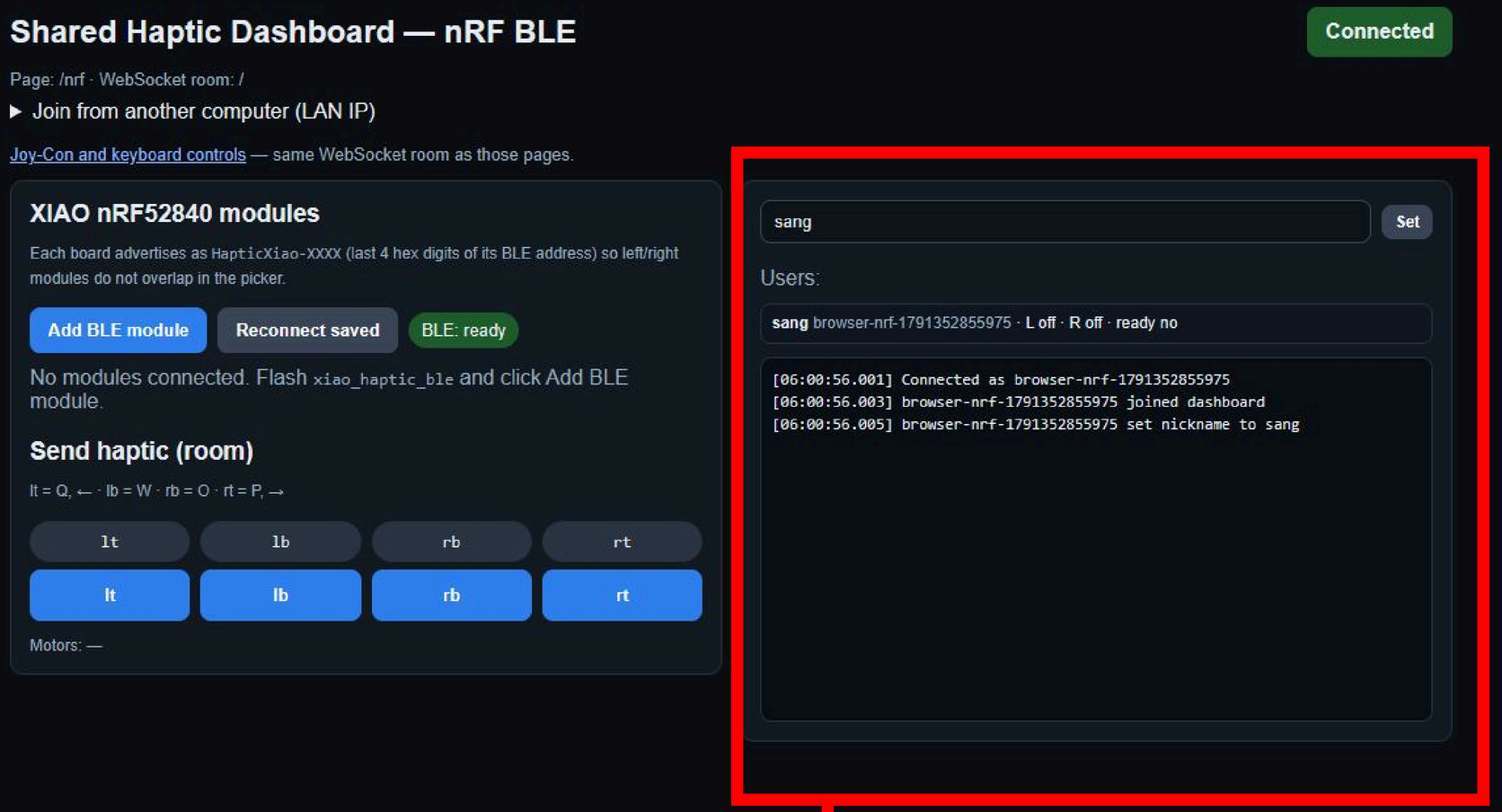
Before connecting device, connect it to your PC’s Bluetooth

Working Web UI



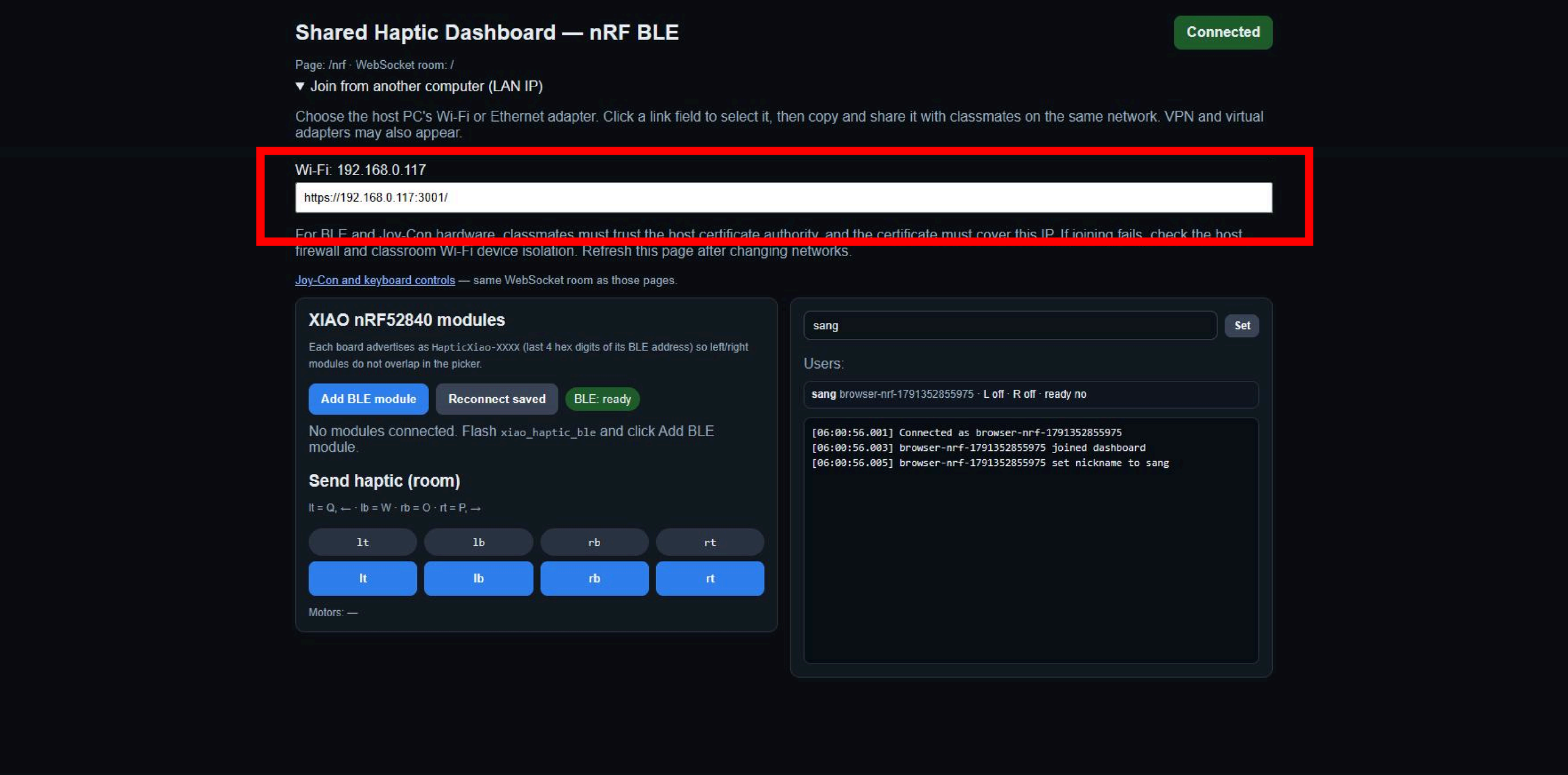
You can use keyboard or buttons to send haptic events to everyone on the same page

Working Web UI



Dashboard for all events

Working Web UI



You can connect to others via browser!